What is UNIX? (And a little bit of history)

The UNIX operating system began life in 1969, in Bell Labs, a division of the American telephone firm, AT&T. There are now many different types of UNIX, making it one of the longest running commercial operating systems available, way longer than Microsoft Windows or Apple MacOS.

UNIX History Timeline

Linux is just one type of UNIX which is most famously known for being an open source, free (as in free speech, not always cost), derivative of UNIX. Most of UNIX's different flavours are still being updated and are still in use all over the world today. Most of the successful ones are based on the AT&T System V (system five) release, which set a standard for UNIX back in 1983. Here are just some popular manufacturers and brands of UNIX, that you may or may not have heard of before:

Developed from 1993, by Sun Microsystems (acquired by Oracle), Solaris was a leader in the commercial UNIX world until the prevalence of open source software & Linux. Solaris still exists today, but Oracle seem less interested in the future of Solaris, and so many organizations have, or are in the process of moving to Linux.

HP's implementation of the UNIX standard System V, called HP-UX was released in 1984 and is still being used today in many enterprise environments.

Which band of big names wouldn't be complete without big blue? Interestingly enough the recent logo of AIX is now green.... Anyway, IBM released their take on Enterprise Unix in 1986, dedicated for it's own IBM hardware, so that they didn't miss out on all of this UNIX malarkey.

AIX actually represents a series of proprietary Unix operating systems. Originally released for the IBM RT PC RISC workstation, it now supports a vast array of different hardware platforms.

Like HP-UX, AIX is based on UNIX System V with 4.3BSD-compatible extensions.

Berkley University: NetBSD and FreeBSD. Berkely Systems Distribution (or BSD) is the closest match to Linux in terms of a direct relationship. Apple macOS and iOS is even built on a modified BSD core (kernel) called Mach. FreeBSD is a fork of BSD which is cost zero.

Pony Up

With the exception of FreeBSD, there was (and still is) a pretty grand fee to own one of the above versions of UNIX. Mainly large commercial organizations and universities have traditionally used these UNIX variants, however Linux appears to be replacing traditional UNIX on a lot of corporate systems due to it's proven track record, it's growing reputation as a contender to UNIX, and it's low price tag, which can often be free.

UNIX is good because it is a true ***multi-tasking***, ***multi-user*** operating system. This means that it can do more than one thing at a time (for example, have a word processor and a music player open and working at the same time) and it can provide all it's services to lots of users at the same time. Modern day workplaces rely on servers to provide a central resource of information and connectivity to users. UNIX was also the platform that many firsts came on: The Internet, the C programming language which is the basis for most modern computer programming languages. These were all firsts that took the other operating systems like Windows and Mac OS a long time to catch up to.

So, Unix is pretty clever, huh? Well, yes. It is, but Unix was also traditionally a pretty boring system that involved learning lots of commands that were tedious to learn.

Why don’t we all use UNIX today if it’s so good?

In 1981, a small company based in Seattle called Microsoft released an operating system. Through chance (Digital Research were supposed to get the contract), IBM took them on to provide an operating system on their new home/small office based computer. This was the IBM PC (or Personal Computer). This operating system was also not graphical. It required commands, in a similar format to UNIX or CP/M, but they were less powerful. The main pitfall of MS-DOS – (Microsoft’s PC Operating System) was, that it had no multi-user, multi-tasking or networking support as standard. By the early 1990’s, this was really starting to wear on PC users. UNIX still had far more power than most operating systems of the time, it was just way too expensive, and legal issues between UNIX vendors licensing UNIX was causing headaches and therefore did not have much exposure outside of large organisations, educational establishments and government offices. It made a lot of sense for most small and medium sized businesses to continue using MS-DOS (and later Windows), it ran the software most people needed, even if it wasn’t delivering the benefits that we all would later take for granted.

During the 80’s, Apple had released another computer, which was separate from the PC, and did not run any PC software, because it relied on it’s own O/S, named MacOS. This time, Apple had decided to make an operating system that was graphical, and later, incorporated colour, pictures, icons and even sounds! Instead of typing everything into the keyboard as commands, the same actions could be made as clicks and movements with a mouse. As with all things Apple, this was revolutionary at the time and changed the face of the world of computing, but still, they hadn’t really grasped UNIX’s multi-tasking, multi-user, networked benefits.

At around the same time, the UNIX world got it’s graphical operating system which began creating a graphical front-end to it’s command-line world, it was called X, or ‘The X Window System’.

In 1990, Microsoft eventually released Windows 3.0 (versions 1 and 2 did not sell well). Windows at the time was a 16-bit, single-tasking, single user, graphical interface built on-top of MS-DOS. UNIX still prevailed: it was multi-user, multi-tasking and it worked on 32 or 64-bit platforms.

It took until 1995, with the advent of Microsoft Windows 95 for Windows to finally go 32 bit, multi tasking. It was also sort-of capable of being multi-user, however it was not best suited: Windows NT came along shortly after, to do that job.